

Discursive Constructions of COIL Virtual Exchange in Institutional Communication: A Corpus-Based Study

Abstract

This article analyzes the discursive recontextualization of Collaborative Online International Learning (COIL) and International Collaborative Projects (PCIs) in the VEm (Virtual Exchange Medium) newsletter of Colleges of Technology within a Brazilian higher education institution. Drawing on a corpus-based approach, keywords were extracted from 33 issues of the newsletter, refined through dispersion criteria, and examined using concordance analysis. The findings reveal three main discursive formations: (1) Institutional Management of COIL Projects; (2) IaH; and (3) Technological Education and Vocational Training. The results indicate that COIL/PCI practices are embedded in a hybrid discursive field in which intercultural and pedagogical ideals of internationalization coexist with managerial, vocational, and organizational rationalities. Within this configuration, Internationalization at Home (IaH), Virtual Exchange, and COIL are recontextualized as PCI practices, which are shaped by institutional logics of management, applicability, and temporal organization within the aforementioned colleges. Results show that virtual internationalization is not a neutral or stable pedagogical model, but rather a discursively mediated practice shaped by local institutional rationalities that simultaneously enable, reconfigure, and delimit its meanings and modes of implementation.

Keywords: Internationalization at Home; Virtual Exchange; Technological Higher Education

Introduction

Higher education internationalization has increasingly shifted beyond physical mobility toward more inclusive and technology-mediated approaches, such as IaH (Beelen & Jones, 2015) and Collaborative Online International Learning (COIL) (Hackett et al., 2024; Leask, 2015; Rubin & Guth, 2015; Succi Junior, 2020). In Brazilian public technological higher education, these initiatives are particularly relevant, as many students face financial, professional, and personal barriers to participating in traditional mobility programs. Within the colleges' network, International Collaborative Projects (PCIs), the institution's local adaptation of COIL, have become an important strategy for promoting intercultural engagement and global learning through virtual collaboration.

While COIL has been widely adopted in higher education, less attention has been paid to how such initiatives are discursively constructed and legitimized within institutional contexts. Internationalization is not only enacted through educational practices but also shaped by the ways it is represented. In this regard, the Virtual Exchange Medium (VEm), a bimonthly newsletter published in Portuguese and dedicated, since 2020, to reporting PCI activities across the *Fatecs* system, provides a valuable site for examining how virtual internationalization is framed and institutionalized.

Using a Corpus Linguistics-informed approach, this study investigates how PCIs/COIL are discursively recontextualized in the VEm corpus, following these research questions:

1. How are PCIs, based on the COIL approach, discursively represented in the VEm newsletter?
2. What dominant discourses emerge in the institutional representation of COIL initiatives within the Fatecs system?
3. How do these discourses recontextualize the meanings and purposes of IaH in the context of technological higher education?

By addressing these questions, the study contributes to discussions on higher education IaH by showing how COIL is adapted, interpreted, and legitimized within the specific context of Brazilian public technological higher education.

Theoretical Background

IaH

The concept of IaH (IaH) was first introduced by Nilsson (2003) and later refined by Beelen and Jones (2015) as the purposeful integration of international, intercultural, and global dimensions into the formal and informal curriculum within domestic learning environments. What stands out in this definition are the principles of intentionality and inclusivity. Internationalization is not understood as incidental exposure to international content but as a structured educational strategy designed to ensure that all students, regardless of their capacity to participate in physical mobility programs, can engage with international and intercultural learning opportunities.

Within contemporary higher education, IaH has increasingly been positioned as a response to the limitations of mobility-centered models of internationalization. As scholars have argued (de Wit & Hunter, 2015; Leask, 2015), international experiences abroad tend to benefit a relatively small proportion of students and frequently reproduce socioeconomic inequalities. In this context, IaH has emerged as a more democratic and accessible approach to internationalization.

This concern is particularly relevant in the context of the colleges, where a substantial proportion of students combine academic studies with full-time employment and most students come from lower- and middle-income households (Centro Paula Souza, 2024). For these students, participation in traditional mobility programs is often financially or logistically unfeasible. Consequently, institutional internationalization strategies have increasingly focused on IaH, which brings international and intercultural experiences into the curriculum without requiring physical mobility. Virtual Exchange (VE) has emerged as one of the primary means of operationalizing IaH, enabling technology-mediated collaboration across borders within domestic educational contexts.

Among the various approaches associated with IaH, VE has gained particular prominence. Virtual Exchange refers to sustained, technology-mediated interaction among participants from different cultural and national backgrounds, organized around collaborative learning objectives (O'Dowd, 2018). Through structured online collaboration, students can develop intercultural, linguistic, and professional competencies without the need for physical mobility.

Within Centro Paula Souza, Virtual Exchange has been operationalized through the International Collaborative Projects, which connect students and faculty from Brazilian Fatecs with international partners through collaborative academic activities conducted online. As a result, PCIs have become one of the institution's primary initiatives for implementing IaH at scale (Succi Junior, 2020).

As the number of PCIs has expanded, so too has the production of institutional documents designed to communicate, regulate, promote, and evaluate these initiatives. These texts constitute more than administrative records. They are institutional discourses through which virtual exchange is represented, legitimized, and operationalized.

COIL and Virtual Exchange: Between Telecollaboration and Intercultural Pedagogy

COIL is situated within an interdisciplinary field that brings together contributions from Applied Linguistics, intercultural pedagogy, and higher education internationalization studies. Although it was formally established as a methodology at the State University of New York (SUNY) in 2006 under the leadership of Jon Rubin, its theoretical foundations can be traced back to earlier traditions of virtual exchange and technology-mediated collaborative learning.

One of the main roots of COIL can be found in telecollaboration, a field that emerged in the 1990s through the work of Applied Linguistics researchers interested in understanding how digital environments could foster meaningful interactions among speakers of different languages and cultures. Initiatives such as *e-tandem*, *teletandem* (Telles & Vassallo, 2006), and other forms of online exchange were based on the premise that technology-mediated interaction promotes not only language learning but also the development of cultural and identity-related competencies (Belz, 2003; O'Dowd, 2007). From this perspective, intercultural communication is no longer viewed merely as an exchange of information; instead, it is conceived as a space for negotiating meanings and co-constructing knowledge.

At the same time, the development of COIL is closely connected to the tradition of intercultural pedagogy. Byram (1997) argues that intercultural learning involves more than the acquisition of knowledge about other cultures; it entails the development of intercultural competence that fosters critical reflection on one's own culture and on the relationships among different social groups. Kramsch's later work shifts the focus from the metaphor of a *third place* to the development of symbolic competence, understood as the ability to interpret, negotiate, and critically engage with multiple symbolic systems and discourses. This per-

spective conceives interculturality as a dynamic process in which multilingual subjects construct fluid identities while navigating diverse linguistic, cultural, and institutional contexts (Kramsch, 2009, 2011).

It was within this theoretical context that Rubin (2017) developed the COIL model, whose main innovation was to move international collaboration beyond the specific domain of language education and into the curriculum as a whole. What COIL proposes is that instructors from different countries co-design academic activities integrated into regular courses across a wide range of disciplines, including engineering, health sciences, business, gastronomy, and technology. COIL is primarily oriented toward the development of intercultural and digital competencies, promoting collaborative online learning across international educational contexts.

According to O'Dowd (2018), Virtual Exchange includes initiatives such as telecollaboration, teletandem, e-tandem, and COIL itself, all of which share key characteristics, including technology-mediated interaction, intercultural engagement, and pedagogical intentionality.

The growth of COIL has also been closely linked to the expansion of the *IaH* agenda. According to Beelen and Jones (2015), internationalization should not rely exclusively on physical mobility, given that only a small proportion of students have access to international study-abroad experiences. Consequently, the COIL model has become strongly associated with principles of inclusion, democratization of access, and the expansion of internationalization opportunities (De Wit & Hunter, 2015; Leask, 2015).

More recently, the literature has emphasized COIL's contribution to the development of global and professional competencies. Hackett et al. (2024) identify four constitutive dimensions of the approach: pedagogical intentionality, intercultural collaboration, technological mediation, and curricular integration. From this perspective, COIL is often described as a tool that fosters competencies valued in contemporary professional contexts, such as teamwork in multicultural settings, collaborative problem-solving, digital literacy, and intercultural communication. This trend is particularly evident in technological and vocational education institutions, where internationalization is frequently aligned with employability goals and preparation for globalized job markets.

However, the growth and institutionalization of COIL have also raised critical concerns. Stein (2018) warns that interculturality may be reduced to a professional competence, thereby emptying its critical and transformative potential. Other studies highlight that the discourse of democratization often obscures inequalities related to technological access, linguistic proficiency, and institutional conditions of participation (O'Dowd, 2021). In addition, there is an increasing alignment between internationalization, performance indicators, and employability agendas, bringing COIL closer to discourses characteristic of contemporary higher education management.

This study addresses a persistent and under-theorized gap in the COIL and

virtual internationalization literature. Despite the rapid expansion of research on COIL over the past decade, the field remains predominantly oriented toward evaluative and outcome-based perspectives, particularly those concerned with pedagogical effectiveness, student engagement, and the development of intercultural competencies. As a result, COIL is still largely conceptualized as a pedagogical innovation instead of a discursively constructed and institutionally mediated phenomenon.

This dominant orientation has left relatively unexplored the ways in which COIL is translated, reframed, and recontextualized as it moves across institutional settings, policy environments, and local educational regimes. Instead of being simply adopted, COIL is discursively reconstructed, acquiring new meanings, functions, and organizational purposes that reflect the rationalities of the institutions in which it is implemented. In fact, there are not enough studies that examine the discursive conditions under which COIL is made intelligible, operationalized, and legitimized within specific institutional contexts, especially in non-elite and vocationally oriented higher education systems in the Global South.

By shifting analytical attention to this issue, the present study changes the focus from COIL as an educational intervention to COIL as a discursive and institutional object, examining how it is constructed and negotiated within this Brazilian public technological higher education.

Internationalization of Public Technological Higher Education in Fatecs

The Colleges of Technology in the state of São Paulo, Brazil, form a public higher education network serving a predominantly working student population, most of whom come from low-income households, with family incomes generally below five minimum wages (Centro Paula Souza, 2024). These socioeconomic conditions significantly limit opportunities for physical mobility, making international experiences inaccessible to many students. Within this setting, IaH provides an alternative pathway for expanding access to international and intercultural learning.

At the same time, the colleges present a distinctive feature within Brazilian higher education: the compulsory inclusion of English and Spanish language courses in the curricula of some programs. This institutional arrangement creates structural affordances for the development of internationalization initiatives. Among these initiatives, PCIs stand out as practices of virtual exchange mediated by international networks and supported by institutional policies for faculty development, pedagogical implementation, and monitoring.

The consolidation of PCIs has not followed a linear trajectory; it has resulted from a gradual process of institutionalization that ranges from isolated initiatives

led by pioneering faculty members to their incorporation into formal governance structures within the institution. This movement entails a shift from localized pedagogical practices to an institutional architecture of internationalization, mediated by administrative coordination, international collaboration networks, and formal monitoring instruments.

Within this context, the VEm newsletter constitutes a privileged site for the circulation and stabilization of institutional discourse on internationalization. It does not function merely as an informational channel, but operates as an institutional device that constructs narratives of internationalization, innovation, and pedagogical transformation.

This framework gives rise to a significant analytical tension: to what extent does the discourse of internationalization circulated in the VEm newsletter both reflect and reconfigure the material and institutional conditions of PCI implementation, while amplifying or performing institutional narratives of internationalization? This study therefore seeks to investigate the relationship between material conditions, institutional practices, and the discursive construction of internationalization within the context of these colleges.

VEm – Virtual Exchange Medium and the Recontextualization of Virtual Exchange Practices

The VEm newsletter is a bimonthly institutional publication produced by the technical unit for Support for the Internationalization of Higher Education (AIES), within the Division of Extension and Research in Higher Education (Depes) and the General Coordination of Undergraduate Higher Education (CGESG) at the institution. Within this institutional architecture, International Collaborative Projects (PCIs) are formally situated within AIES/Depes/CGESG, reflecting their embeddedness in a highly stratified administrative structure.

PCIs constitute the institution's local implementation of IaH, an approach that seeks to provide international and intercultural learning opportunities without requiring physical mobility. One of the principal strategies for achieving this objective is VE, which promotes technology-mediated interaction among participants from different countries. "Collaborative Online International Learning (COIL) represents a structured model of VE in which international partner instructors co-design and implement collaborative projects grounded in, but not limited to, their respective course disciplines." At the institution, these principles have been adapted into the PCI model, a localized form of COIL and VE designed to promote IaH across the Fatecs network.

Although PCIs are internationally recognized as COIL or Virtual Exchange initiatives, their implementation within the colleges involves a process of contextual adaptation often referred to as "tropicalization" (Succi 2020). This process raises important analytical questions regarding how globally circulating models of internationalization are recontextualized within local institutional settings and how such recontextualization may transform both the meaning and the function of these pedagogical approaches.

Framed within discourses of global citizenship education and IaH, PCIs are presented as enabling students to engage in collaborative academic activities aligned with curricular content while fostering digital, linguistic, intercultural, and teamwork competencies. However, this framing also invites critical examination of how such competencies are discursively constructed as evidence of institutional internationalization and how these narratives circulate through official communication channels such as the VEm newsletter.

This configuration gives rise to an important analytical question: how does the institutional discourse surrounding PCIs negotiate the tension between globally circulating models of virtual exchange and their localized reconfiguration within a highly structured administrative framework, thereby shaping the discursive construction of internationalization in the VEm newsletter?

Corpus Linguistics

Corpus Linguistics is an area within Applied Linguistics concerned with the analysis of corpora, understood as systematically compiled collections of spoken and written texts used for linguistic research (Berber Sardinha, 2004). While corpora are often defined as representing “authentic” language use produced by real speakers, such notion requires critical scrutiny, particularly in institutional contexts where texts are highly mediated, edited, and shaped by organizational purposes. In this study, VEm newsletters are treated as a corpus, not as a transparent reflection of linguistic reality, but as a structured site of institutional discourse production.

The development of Corpus Linguistics has been closely associated with the empirical turn in linguistics and the rejection of purely introspective methods. However, the assumption that computational tools merely enable more efficient analysis overlooks the fact that they also participate in shaping what counts as analyzable linguistic evidence. Word lists, concordancers, and other tools do not simply reveal patterns; they also construct them through specific operationalizations of language data.

Corpus Linguistics is often described as adopting an empiricist and probabilistic view of language, in which linguistic patterns are considered systematic and statistically observable. However, such a perspective raises methodological questions regarding the relationship between frequency and meaning, as well as the extent to which distributional patterns can be directly interpreted as social or ideological regularities.

The integration of Corpus Linguistics with Critical Discourse Analysis (CDA) (Baker et al., 2008) has been proposed as a way of addressing the limitations of each approach. While CDA provides tools for interpreting discourse within its social and ideological contexts, it has often been criticized for limited datasets and selective textual evidence. Conversely, corpus-based approaches enable the analysis of larger datasets, but risk reducing discourse to quantifiable patterns if not theoretically contextualized. What emerges is not a simple integration

but a productive epistemological negotiation between approaches, especially in relation to scale, representation, and interpretation.

In this study, we adopt a corpus-assisted CDA approach to analyze 33 editions of the VEm newsletter (70,933 words). However, this corpus is not understood as a statistically representative sample of language use in general, but as a bounded institutional archive through which discourses of internationalization are produced, circulated, and stabilized over time. Consequently, methodological decisions regarding corpus design, inclusion criteria, and analytical procedures are not neutral, but directly implicated in the construction of the object of study (Biber et al., 1998).

Methodology

The VEm corpus in this study was compiled by collecting all VEm newsletters published from June 2020 to February 2026 (issues 1-33) and downloaded from the website <https://publicacoescgesg.cps.sp.gov.br/VEm/issue/archive>. Each edition was saved as a PDF file and named numerically according to its issue number. For example, issue 1 was named VEm_01.

We concluded that 33 files would account for a relatively small number for a discourse analysis based on Corpus Linguistics (McEnery & Hardie, 2012). Therefore, we decided to segment each edition into separate files according to the different texts. As a result, each text became a single file, totaling 180 files. The corpus can be seen in Table 1.

Table 1 – The VEm corpus

Texts	Unique Words	Total Words
180	7,873	70,933

Source: prepared by the authors

The textual data analyzed in this study consisted of PDF documents whose embedded text layers did not reliably preserve the original reading order, as the files had been created using page-layout software. Textual elements were often fragmented or interleaved, making direct extraction unsuitable for corpus analysis. To address this issue, a multi-stage reconstruction and verification pipeline was developed.

First, each PDF was divided into individual pages, which were processed as separate units. For every page, two representations were generated: a conventional text extraction from the PDF and a high-resolution image for image-based optical character recognition (OCR). In the second stage, the page images were processed using ChatGPT’s 5.2 Pro vision-based OCR capabilities. A structured prompt guided the reconstruction of the page content, and the output was returned in JSON format containing titles, headings, and paragraph-level text. To ensure that the corpus contained only communicative content, recurring layout

elements such as mastheads, footers, captions, and other non-prose artifacts were excluded from the extraction process.

Given the well-known possibility that large language models may introduce lexical items not present in the original source, an additional verification stage was implemented. The reconstructed OCR text was compared against the reference text extracted directly from the PDF. Token-level comparisons were performed between the two versions of each page. Tokens present in the OCR reconstruction but absent in the PDF text were flagged as potential hallucinations or recognition errors. For example, in one instance the OCR system produced *aumentado* ('increased') where the original token was *aumentando* ('increasing'). Such discrepancies typically arise from ambiguity in character recognition (e.g., confusion between visually similar character sequences such as -s and -cs or -do and -ndo). To assist with identifying these discrepancies, normalized Levenshtein distance was used to measure similarity between tokens in the OCR output and those in the reference extraction. In addition, keyword-in-context (KWIC) displays were generated so that each flagged token could be examined within its surrounding textual environment. These contextual displays allowed the researcher to determine whether a token represented a plausible OCR recognition error, a minor orthographic deviation, or a lexical insertion introduced during the reconstruction process. Manual corrections were then recorded and applied to the reconstructed text.

After manual review and correction, the cleaned page-level texts were recombined into complete documents corresponding to the original PDFs. The final corpus was stored in tabular format, enabling subsequent analyses such as frequency, keyword, and collocation studies.

Keyword analysis was used to identify lexically salient items. Instead of comparing the corpus to an external reference corpus, each newsletter issue was treated in turn as a study corpus against a reference corpus composed of the remaining 32 issues. This approach captured patterns of lexical salience within the institutional discourse itself. The procedure generated 33 keyword lists, which were subsequently merged into a single list.

To strengthen analytical reliability, a dispersion filter was applied: only keywords occurring in at least five documents (approximately 3% of the corpus) were retained. This threshold prioritized recurrent lexical patterns while reducing the influence of highly localized occurrences. At the same time, it is recognized that such filtering shapes the analytical object by privileging recurring institutional discourse and excluding low-frequency items that may still hold contextual or ideological significance. After filtering, 23 keywords remained and were used as the analytical variables for the subsequent discourse analysis (see Table 2).

Table 2 – Corpus keywords

Keyword	Issue in which it appears		Log-likelihood
	most frequently	Number of files	
every	vem_ed_08	69	08.90
October	vem_ed_32	33	13.42
hour	vem_ed_11	24	05.19
September	vem_ed_31	20	12.54
operation	vem_ed_02	19	11.45
inclusion	vem_ed_25	13	11.77
energy	vem_ed_19	09	16.09
goal	vem_ed_05	09	13.46
panel	vem_ed_10	09	10.03
technologist	vem_ed_01	08	12.48
less	vem_ed_06	08	11.45
school term	vem_ed_14	08	07.77
equipment	vem_ed_07	07	13.46
poster	vem_ed_30	07	13.35
performance	vem_ed_28	07	12.17
annual	vem_ed_24	07	09.85
free	vem_ed_23	06	17.45
audience	vem_ed_18	06	14.66
logistics	vem_ed_04	06	08.63
medical	vem_ed_02	05	11.45
dedication	vem_ed_22	05	11.40
office	vem_ed_03	05	08.63
reception	vem_ed_03	05	08.63

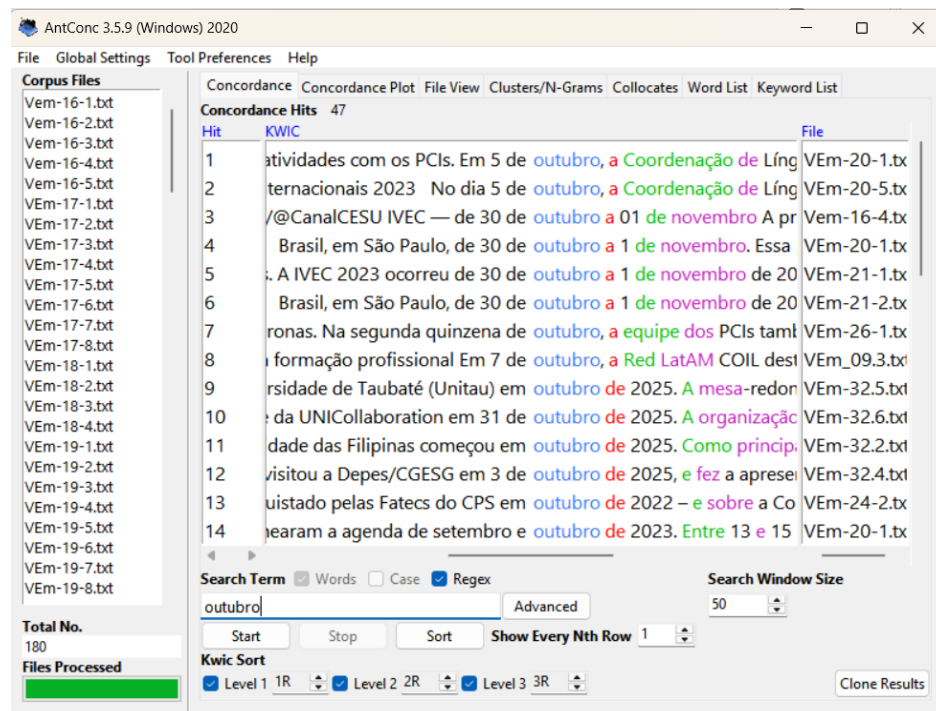
Source: prepared by the authors

Collocational patterns were examined using the log-dice association measure, implemented in AntConc (Anthony, 2020). The log-dice score was used to quantify the strength of association between lexical items within a defined window span, allowing for the identification of statistically salient co-occurrence patterns.

It is important to distinguish between keyword status and collocational strength: while keyword analysis captures distributional salience across documents, collocation analysis captures relational proximity between lexical items within textual environments. These two levels of analysis are treated as complementary but analytically distinct.

To contextualize lexical and collocational findings, concordance analysis was conducted using AntConc’s Concordance and File View tools. Keyword occurrences were examined in their immediate co-text, enabling qualitative interpretation of usage patterns.

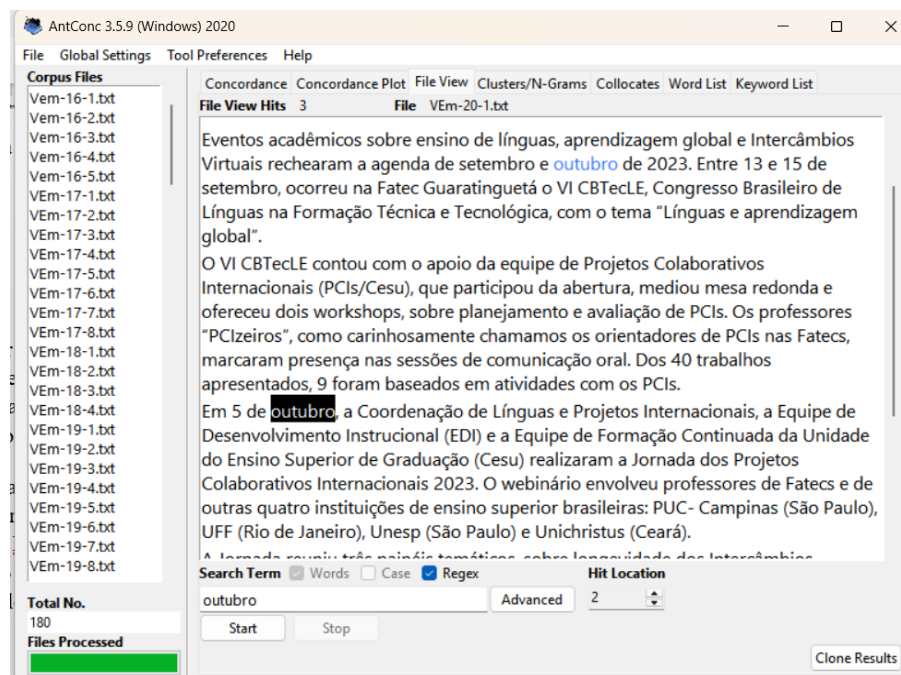
Figure 1 – Files containing the word “October” in the corpus



Source: Prepared by the authors using the Concordance tool in AntConc (2026).

Each concordance line was inspected individually, with extended contextual segments retrieved when necessary to ensure interpretive accuracy. This procedure supports a corpus-assisted qualitative reading, in which statistical patterns are systematically triangulated with contextual discourse analysis.

Figure 2 – Text analyzed based on the selected keyword in the corpus



Source: Prepared by the authors using the File View tool in AntConc (2026).

Following extraction and filtering, the keywords were grouped into three discursive clusters based on their functions and contexts of use across the corpus. This categorization was informed by concordance analysis and refined through iterative interpretation.

The analysis combines quantitative corpus methods with qualitative discourse interpretation within a corpus-assisted discourse studies (CADS) framework. Instead of treating these components as separate stages, the approach views them as mutually constitutive in the production of discourse-analytical findings.

Adopting a corpus-assisted CDA perspective, the study links large-scale pattern identification with context-sensitive interpretation. Corpus linguistics provides systematic evidence of distributional patterns, while CDA enables their critical examination in relation to institutional and ideological contexts. This integration also acknowledges methodological tensions between frequency and meaning, and between distributional regularities and discursive interpretation.

Accordingly, the VEm corpus is not treated as a transparent representation of institutional discourse, but as a structured and mediated site of discourse production in which institutional meanings are constructed, circulated, and stabilized.

The following section presents the results of the corpus analysis, focusing on the identified keywords, their dispersion patterns, and their discursive organization

across the VEm newsletter corpus.

Discussion and Results

The analysis suggests that the VEm newsletter does not merely report PCI/COIL initiatives, but actively participates in the construction of institutional meanings surrounding internationalization. Although COIL is internationally associated with principles such as intercultural dialogue, reciprocity, and collaborative knowledge construction, its representation within the newsletter reflects a process of discursive recontextualization through which global educational models are adapted to local institutional priorities.

This raises an important analytical question: which discourses of COIL are foregrounded, transformed, or marginalized in this process of institutional translation? Instead of appearing as neutral descriptions of practice, the recurring discursive configurations identified in the corpus may be understood as manifestations of broader institutional rationalities that shape how internationalization is imagined, justified, and operationalized within public technological higher education.

The three discursive configurations identified in the corpus (Institutional Management of COIL Projects, IaH, and Technological Education and Vocational Training) should therefore not be viewed merely as thematic categories. Instead, they may be interpreted as competing yet interconnected frameworks through which the meaning and purpose of PCI/COIL are negotiated. Their coexistence suggests that internationalization is simultaneously constructed as an administrative project, a strategy for widening access to global experiences, and a mechanism for workforce preparation.

In this sense, the findings invite reflection not only on how internationalization is represented in institutional discourse, but also on the institutional priorities and assumptions that such representations help legitimize and reproduce.

Discourse 1 - Institutional Management of COIL Projects

The first discursive configuration identified in the VEm corpus is labeled *Institutional Management of COIL Projects*. This label is not treated as an inherent property of the data, but as an interpretive construct derived from recurring distributional patterns identified through keyword analysis, dispersion mapping, and concordance examination.

Across the corpus, a set of lexical items related to planning, temporality, evaluation, and institutional coordination systematically co-occurs in contexts in which Collaborative International Projects (PCIs) and COIL initiatives are described. These include terms such as *goal*, *operation*, *panel*, *annual*, *performance*, *dedication*, *school term*, *October*, and *September*. Their distribution suggests a recurrent alignment between internationalization initiatives and institutional cycles of organization, monitoring, and temporal structuring.

This pattern is particularly visible in procedural descriptions of project activities,

in which COIL initiatives are organized through sequential steps, comparative tasks, and structured phases of implementation. In one instance, student collaboration is framed through operational procedures involving the selection of technical objects, comparative analysis of products across countries, and systematic data collection on functionality and maintenance. Excerpt 1 situates the project within clearly defined temporal phases and iterative redesign across editions of the initiative. The following excerpts were translated, and the words in bold represent this discourse:

Excerpt 1 – In 2019, the student groups (10 Americans and 17 Brazilians) discussed the equipment they were familiar with and compared brands and prices in both countries. Each team selected a piece of equipment and gathered information about its functioning, **operation**, and maintenance. The second edition of the project, held during the second semester of 2020, will address the context of COVID-19: students will discuss the situations experienced in the two countries and, in particular, the protocols for sanitizing medical equipment. (VEm – Issue 02 - News section)

Instead of portraying international collaboration solely as a pedagogical activity, these passages frame it through procedural and managerial logics that emphasize planning, coordination, and institutional control.

A second recurring pattern is the association of COIL/PCI initiatives with strategic planning, goal-setting, and adaptive management. Across the corpus, successful implementation is frequently linked to clearly defined objectives, structured processes, and the capacity to adjust projects when necessary, as illustrated in the Excerpt 2:

Excerpt 2 – Therefore, self-motivation and motivating the class are important. It is also essential to have a project with clear **goals** and a roadmap with defined deadlines. This is because, even with planning, adjustments are necessary. (VEm – Issue 5 - “Best Practices” section)

These formulations present internationalization as a planned and measurable process aligned with institutional goals and organizational expectations. A third recurring pattern is the use of quantitative indicators and evaluative metrics. Numerical data on participation, performance, and external recognition are frequently mobilized to legitimize COIL/PCI initiatives and demonstrate their value, as illustrated in the Excerpt 3:

Excerpt 3 – A piece of news that brings great pride: in July 2022, colleges from the institution stood out among the institutions with the highest number of COIL projects (Virtual Exchanges) in the world, according to the COIL Connect for Virtual Exchange website (<https://coilconnect.org/>). Last year alone (August 2021–July 2022, following the American **school term** as a reference), there were 109 PCIs: 39 in the second semester of 2021 and 70 projects in the first semester of 2022. The perception of students and professors involved in PCIs is highly positive, as shown on pages 4 and 5. PCIs have also generated academic publications and participation in international conferences. (VEm – Issue 14 - Editor’s Letter section)

In this configuration, numerical data function not simply as descriptive information but as resources for institutional legitimacy and visibility. This pattern is reinforced through references to certifications, awards, and external forms of recognition that validate PCI/COIL initiatives. The attribution of the FAUBAI-BRaVE seal, for example, is presented as evidence of program quality and institutional achievement:

Excerpt 4 – In **October** 2022, the institution received the FAUBAI-BRaVE seal from the Brazilian Association for International Education (FAUBAI). This was one of the main achievements of the Collaborative International Projects team (PCIs) that year. BRaVE is an acronym for Brazilian Virtual Exchange, and the awarding of the seal to CPS attests to the quality of PCIs. (VEm – Issue 16 - - Editor’s Letter section)

Such references construct internationalization as a practice legitimized through external recognition. A further recurring pattern is the integration of COIL/PCI initiatives into institutional schedules and academic events, as illustrated in Excerpt 5:

Excerpt 5 – Academic events on language teaching, global learning, and Virtual Exchanges filled the agenda of **September** and **October** 2023. Between **September** 13 and 15, the VI CBTEcLE (Brazilian Congress on Languages in Technical and Technological Education) was held at Fatec Guaratinguetá, with the theme “Languages and global learning.” (VEm – Issue 20 - Editor’s Letter section)

This temporal framing reinforces the integration of internationalization into institutional routines and the rhythms of academic governance. Taken together, these patterns suggest that COIL/PCI initiatives are articulated through a discourse of institutional organization, in which planning, evaluation, recognition, and temporality play important roles. However, this discourse is not

homogeneous. Managerial, pedagogical, and internationalization logics coexist in different configurations across the corpus, depending on the genre, issue, and communicative purpose.

From this perspective, the label *Institutional Management of COIL Projects* is understood as a corpus-derived analytical construct emerging from recurring lexical, temporal, and evaluative patterns instead of an externally imposed theoretical category. Its prominence is further reinforced by the editorial architecture of the newsletter itself, which includes intentionally designed sections that articulate institutional perspectives (e.g., Editor’s Letter) and highlight “Best Practices” as models for the implementation of future COIL initiatives.

At the same time, the analysis reveals a constitutive tension in the representation of PCIs/COIL initiatives. While grounded in intercultural collaboration and pedagogical exchange, these initiatives are also embedded in institutional frameworks that emphasize planning, measurability, scalability, and accountability. This tension does not eliminate the pedagogical dimension of COIL but shapes how internationalization becomes visible, manageable, and communicable within the VEm newsletter.

Discourse 2 - IaH

The second axis, IaH, encompasses discourses that highlight the collaborative, inclusive, and relational dimensions of PCIs/COIL. Internationalization is portrayed as an accessible experience, enabled by digital technologies and supported by global networks of cooperation. Diversity, expanded access, and the engagement of faculty and students emerge as central elements, aligning the institutional discourse with the core principles of IaH.

Keywords: inclusion, free, audience, reception, office, poster, hour, every, less

Excerpt 6 – The theme of the 4th Red LatAM COIL Conference was “diversity, equity and **inclusion**.” A professor, from our PCIs team, is a member of the network’s executive council and participated in the opening of the event on September 4. On the 5th, she presented a workshop with colleagues from a university in Venezuela and The United Kingdom). (VEm - Issue 25 - Editor’s Letter section)

An open and connected model of international education grounded in collaboration, expanded access, and the global circulation of knowledge.

Core:

The democratization of internationalization through diversity and global learning communities. Education is conceived as a transnational space of collaboration, although inequalities in access are often obscured by institutional discourse.

Excerpt 7 – She mentioned the phases of PCIs developed at the colleges: icebreaker, intercultural moment, main task, “deliverables,” presentation, reflection, and assessment. She also reported three PCIs carried out with Colombia, focusing on slang in different communicative contexts, business advertising, and digital advertising.

When asked by the **audience** about how to design an effective project so that students develop not only language skills but also empathy and intercultural understanding in an authentic way, Regiane replied: “It is necessary to carefully plan the icebreaker and break stereotypes about the participating countries, guiding students toward clear intercultural communication, with respect for political and religious differences in order to avoid discomfort.” (VEm – Issue 31 - News section)

The analysis of the excerpts shows that the VEm newsletter constructs COIL/PCI as an internationalization practice grounded in openness, collaboration, and diversity, closely aligned with the core principles of the COIL model. The reported experiences emphasize intercultural spaces of exchange, such as workshops, international networks, and collaborative environments, where students and faculty share perspectives, experiences, and knowledge on a global scale.

Excerpt 8 – On August 29, Fatec Indaiatuba hosted a virtual **reception** for Foreign Trade students, bringing together professors from institutions in four countries, including Brazil. Entitled “Higher Education and COVID-19: A Global Issue,” the event was moderated by the program coordinator, and featured the participation of the campus director. (VEm – Issue 03 - “Best Practices” section)

This discourse aligns closely with the original COIL model, which frames COIL as a means of democratizing internationalization. However, it also carries an implicit limitation: unequal access to technology persists, and partnerships still rely on institutional networks.

Internationalization is no longer tied exclusively to physical mobility but is instead conceived as an accessible, distributed, and inclusive experience mediated by digital technologies. This aligns with COIL literature, which defines the approach as promoting intercultural learning and collaboration across countries without physical travel, thereby expanding access to internationalization (Rubin & Guth, 2015).

Excerpt 9 – During the first meeting, on September 18, the 26 Brazilian students used Zoom to ask the American professor about her professional experience, adaptation to online teaching, and employability during the pandemic. To prepare the students, the professor offers English conversation workshops in addition to the two weekly **hours** of English instruction required in the college curriculum, a distinguishing feature compared to other institutions of technological higher education. According to the professor, “the project is important for students not only because it helps improve their English skills, but also because it provides contact with students from another country and the opportunity to discuss issues of global interest.” (VEm – Issue 04 - “Best Practices” section)

The recurrence of terms such as *inclusion*, *free*, *audience*, and *reception* constructs education as an open global network in which knowledge circulates in a decentralized and participatory way. Within this framing, COIL/PCI is represented as an expanded mechanism for international experience beyond traditional mobility programs.

However, this discourse also abstracts from unequal conditions of access. By emphasizing openness and intercultural exchange, it tends to portray participation as universally available while downplaying technological, linguistic, institutional, and geopolitical asymmetries. As a result, it constructs an idealized vision of global educational participation.

Overall, the IaH discourse plays a dual role in the VEm corpus: it legitimizes COIL through values such as inclusion and diversity, while also promoting a universalized narrative that partially obscures structural inequalities, simultaneously democratizing and depoliticizing internationalization.

Discourse 3 - Technological Education and Vocational Training

The third axis, Technological Education and Vocational Training, recontextualizes COIL/PCI as an instrument for internationally oriented technical qualification. Intercultural collaboration is primarily mobilized to develop work-related competencies, including technical, digital, collaborative, and intercultural skills. PCIs are thus integrated into the colleges’ educational mission as a strategic resource for training technologists, linking pedagogical innovation with employability.

Keywords: technologist, equipment, energy, medical, logistics

Excerpt 10 – The collaboration began in the second semester of 2019; each edition involved approximately 50 students (25 from each country) and five professors, one from an American university and four from one Brazilian campus: participants included the professors of International Marketing, International Business and English. According to the course coordinator, the initiative helps future **technologists** develop “intercultural and interpersonal competencies, the ability to work in mixed teams, solve problems, and use technologies.” (VEm – Issue 1 - News section)

The emphasis on teamwork, problem-solving, technology use, and intercultural competencies frames internationalization less as cultural engagement and more as a tool for developing skills valued in contemporary job markets. Interculturality is thus positioned less as an educational end in itself and more as a professional competency linked to future employability.

The valorization of practical and technological education focuses on applicability, technical innovation, and practice-oriented academic production, with education understood as training for technical performance in the real world.

Excerpt 11 – The **Energy** and Environment Professor at one of the colleges. From an environmental perspective, the PCI was fundamental for joint learning about the notion of “thinking globally” — since the search for sustainable energy alternatives, in the face of the challenges posed by climate change, transcends geographical, political, or cultural barriers — and “acting locally.” (VEm – Issue 19 - “Best Practices” section)

The examples reinforce an applied conception of internationalization in which intercultural collaboration is tied to practical outcomes and professional problem-solving, whether in sustainability challenges or real business cases. PCIs/COIL is thus situated within a competency-based education framework, where its value lies in contributing to innovation, technical training, and workforce preparation.

Excerpt 12 – In the first semester, after an initial meeting with several representatives from Inacap (Chile), collaborations began. In the second semester, a series of **logistics** webinars involved professors from 2 colleges. In 2021, one Challenge involved 17 mixed teams of Brazilian and Chilean students, who worked together to solve a real case from a company in the pet sector called Viejito Perro. (VEm – Issue 9 - News section)

The analysis shows that the VEm newsletter constructs PCIs/COIL as a practice-oriented professional training tool aligned with technological education. The

reported experiences emphasize competencies such as teamwork, problem-solving, and technology use, framing internationalization not only as an intercultural experience but also as a strategic resource for technical qualification.

This discourse shifts the emphasis of COIL from intercultural engagement as an end in itself toward its instrumental value in developing professional competencies, thereby aligning internationalization with the vocational mission of the colleges. The emphasis on technological programs, interdisciplinary articulation (e.g., marketing, business, and English), and evaluation tools highlights concerns with effectiveness and continuous improvement, aligning COIL with evidence-based pedagogical innovation.

PCIs/COIL is also framed as part of an ecosystem of applied academic production, where theory and practice are integrated in real or simulated professional contexts. Education is thus conceived as direct preparation for technical work, closely aligned with contemporary demands for employability and global competence.

The recurrent references to project management, technological learning, academic outputs, and measurable outcomes indicate a growing alignment between internationalization and performance-oriented educational practices. COIL success is often assessed through tangible results, skills acquisition, and institutional performance, reflecting a managerial view of educational innovation.

Overall, internationalization is increasingly subordinated to workforce preparation. Instead of being a space for critical engagement with cultural difference or global inequalities, PCIs/COIL is mainly framed as an initiative for producing job-relevant competencies, which reflects broader neoliberal shifts in higher education, where internationalization is legitimized through employability, innovation, and competitiveness. Interculturality is thus redefined as a transferable skill, with cultural encounters functioning as resources for human capital development, which narrows its critical and emancipatory potential.

Taken together, the three axes reveal a hybrid discursive field in VEm, where intercultural pedagogical ideals coexist and often tension with managerial logics and applied training goals. These rationalities overlap and sometimes conflict in the construction of IaH at the colleges.

Conclusions

The discursive analysis of the VEm newsletter reveals that International Collaborative Projects (PCIs), grounded in the COIL approach, are constructed through three interconnected discourses: Institutional Management of COIL Projects, IaH, and Technological Education and Vocational Training. Together, these discourses shape how virtual internationalization is understood and legitimized within the Fatecs system.

The first discourse frames COIL as a governable educational intervention organized through planning, monitoring, evaluation, and institutional coordination. This managerial logic supports the institutionalization, visibility, and scalability

of IaH initiatives while aligning internationalization with broader principles of organizational management and accountability.

The second discourse reflects the normative foundations of COIL and IaH by representing internationalization as an inclusive, collaborative, and digitally mediated experience. Diversity, inclusion, and intercultural exchange are important values, positioning virtual internationalization as a democratizing force. However, this representation tends to obscure inequalities related to technological access, linguistic capital, and institutional resources, presenting participation as more universally accessible than it may be in practice.

The third discourse recontextualizes COIL as a mechanism for developing competencies valued in global labor markets. Intercultural collaboration is primarily legitimized through its contribution to technical, digital, collaborative, and intercultural skills, reinforcing the vocational orientation of technological higher education. As a result, interculturality is increasingly represented as professional capital instead of transformative educational experience.

Taken together, these discourses reveal a hybrid formation in which pedagogical, managerial, and vocational rationalities coexist. Yet managerial and employability-oriented logics increasingly function as the primary frameworks through which internationalization is legitimized. Intercultural collaboration acquires institutional value largely through its contribution to measurable outcomes, organizational efficiency, and professional qualification.

The findings suggest that IaH is not a neutral or universally enacted process but a context-dependent practice shaped by institutional priorities and competing organizational rationalities. Instead of being transferred unchanged across settings, COIL is continuously reconfigured according to local governance structures, educational missions, and professional training goals.

More broadly, the study highlights how contemporary internationalization is increasingly expected to be inclusive, measurable, scalable, and economically relevant simultaneously. While these objectives may coexist, they also generate tensions. The analysis therefore raises important questions about what is transformed, or potentially lost, when internationalization becomes closely linked to governance, performance, and employability. Future research should move beyond assessing whether virtual exchange promotes internationalization and examine more critically which forms of internationalization are being produced and legitimized through these practices.

Recommendations

Based on the findings, institutions implementing COIL and related virtual exchange initiatives should monitor how internationalization is represented in institutional communication alongside how projects are organized and evaluated. Institutional documentation can make explicit the intercultural and pedagogical purposes of virtual exchange while also acknowledging differences in technological access, linguistic resources, and participation conditions. Future institutional

planning may benefit from combining indicators of project implementation and professional learning with evidence of intercultural learning and student experience. For research, longitudinal corpus-based analyses of institutional communication can help examine how the meanings and functions of virtual internationalization change as programs become more established.

Ethics Statement

This study analyzed publicly available institutional newsletters and did not involve the recruitment or interaction of human participants or animal subjects. No personal data were collected from participants for research purposes. Generative artificial intelligence was used during corpus preparation exclusively for page-level optical character recognition (OCR) and textual reconstruction of PDF documents. ChatGPT 5.2 Pro was used to reconstruct text from page images. The reconstructed text was systematically compared with text extracted directly from the original PDFs, and discrepancies were manually examined and corrected by the researchers. The authors take full responsibility for the accuracy and integrity of the resulting corpus and for the interpretation of the findings.

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Conflict of Interest

The authors declare no conflict of interest related to this research or its publication.

References

- Anthony, L. (2020). *AntConc* (Version 3.5.9) [Computer software]. Waseda University. <https://www.laurenceanthony.net/software/AntConc>
- Baker, P., Gabrielatos, C., Khosravini, M., Krzyzanowski, M., McEnery, T., & Wodak, R. (2008). A useful methodological synergy? Combining critical discourse analysis and corpus linguistics to examine discourses of refugees and asylum seekers in the UK press. *Discourse & Society*, 19(3), 273–306.
- Beelen, J., & Jones, E. (2015). Redefining IaH. In A. Curaj, L. Matei, R. Pricopie, J. Salmi, & P. Scott (Eds.), *The European higher education area: Between critical reflections and future policies* (pp. 59–72). Springer.
- Belz, J. A. (2003). Linguistic perspectives on the development of intercultural competence in telecollaboration. *Language Learning & Technology*, 7(2), 68–117. <https://doi.org/10.10125/25201>
- Berber Sardinha, T. (2004). *Linguística de corpus*. Manole.
- Biber, D., Conrad, S., & Reppen, R. (1998). *Corpus linguistics: Investigating language structure and use*. Cambridge University Press.

Byram, M. (1997). *Teaching and assessing intercultural communicative competence*. Multilingual Matters.

Centro Paula Souza. (2024, April 5). *CPS divulga dados do relatório socioeconômico das Fatecs*. <https://www.cps.sp.gov.br/centro-paula-souza-divulga-dados-do-relatorio-socioeconomico-das-fatecs/>

de Wit, H., & Hunter, F. (2015). The future of internationalization of higher education in Europe. *International Higher Education*, 83, 2–3.

Hackett, S., Dawson, M., Janssen, J., & van Tartwijk, J. (2024). Defining Collaborative Online International Learning (COIL) and distinguishing it from Virtual Exchange. *TechTrends*, 68(6), 1078–1094. <https://doi.org/10.1007/s11528-024-01000-w>

Kramsch, C. (2009). *The multilingual subject*. Oxford University Press.

Kramsch, C. (2011). The symbolic dimensions of the intercultural. *Language Teaching*, 44(3), 354–367. <https://doi.org/10.1017/S0261444810000431>

Leask, B. (2015). *Internationalizing the curriculum* (1st ed.). Routledge. <https://doi.org/10.4324/9781315716954>

McEnery, T., & Hardie, A. (2012). *Corpus linguistics: Method, theory and practice*. Cambridge University Press.

Nilsson, B. (2003). Internationalisation at home from a Swedish perspective: The case of Malmö. *Journal of Studies in International Education*, 7(1), 27–40. <https://doi.org/10.1177/1028315302250178>

O'Dowd, R. (2007). Evaluating the outcomes of online intercultural exchange. *ELT Journal*, 61(2), 144–152. <https://doi.org/10.1093/elt/ccm007>

O'Dowd, R. (2018). From telecollaboration to virtual exchange: State-of-the-art and the role of UNICollaboration in moving forward. *Journal of Virtual Exchange*, 1(1), 1–23. <https://doi.org/10.14705/rpnet.2018.jve.1>

O'Dowd, R. (2021). Virtual exchange: Moving forward into the next decade. *Computer Assisted Language Learning*, 34(3), 209–224. <https://doi.org/10.1080/09588221.2021.190220>

Rubin, J. (2017). Embedding Collaborative Online International Learning (COIL) at higher education institutions. *Internationalisation of Higher Education*, 2, 27–44.

Rubin, J., & Guth, S. (2015). Collaborative Online International Learning: An emerging format for internationalizing curricula. In A. S. Moore & S. Simon (Eds.), *Globally networked teaching in the humanities: Theories and practice* (pp. 15–27). Routledge.

Stein, S. (2018). Beyond higher education as we know it: Gesturing towards decolonial horizons of possibility. *Studies in Philosophy and Education*, 38(2), 143–161. <https://doi.org/10.1007/s11217-018-9622-7>

- Succi Junior, O. (2020). Projetos colaborativos internacionais na unidade de ensino superior de graduação: A evolução dos intercâmbios virtuais no Centro Paula Souza. *Revista de Estudos de Gestão, Informação e Tecnologia*, 14, 126–140.
- Telles, J. A., & Vassallo, M. L. (2006). Foreign language learning in-tandem: Tele-tandem as an alternative proposal in CALLT. *The ESPECIALIST*, 27(2), 189–212.
- VEm – Virtual Exchange Medium. (2020–present). *Informativo dos PCIs da CGESG*. <https://publicacoescgesg.cps.sp.gov.br/VEm/>